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Effect of immersion in SBF on porous bioactive bodies made by sintering bioactive glass microspheres

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Abstract

Since the mid 90s it has been possible to draw fibers and manufacture microspheres of novel bioactive glasses. Thus, by sintering bioactive glass microspheres it is now possible to form porous textures, in which the bioactive surface area is increased manifold compared with non-porous bodies. Four different types of porous bodies were made by sintering glass microspheres of diameter 250–300 μm . Two of the body types contained only one kind of spheres; either highly bioactive glass spheres or spheres made of glass having a low bioactivity (biocompatible glass). Two additional types of test bodies were obtained by sintering mixtures of bioactive and biocompatible spheres (composites). The dissolution of silica and calcium into simulated body fluid (SBF) was determined at different time intervals using a direct current plasma atomic emission spectrometer (DCPAES). The influence of immersion on the mechanical strength of the porous structures was studied by means of a compression test. Further, the thickness of the silica-rich gel formation on the surface of bioactive glass spheres was measured at each time interval using back-scattered electron imaging of scanning electron microscopy (BEI-SEM). A non-porous glass rod made from the same bioactive glass was used as the control. The results showed that dissolution of silica and calcium into SBF from the porous glass texture was inversely related to the silica content of the glass. The rate of silica gel formation on the sintered bioactive microspheres was significantly higher than on a rod made from the same glass. The initial mechanical strength of porous bodies consisting of only one kind of glass was 17–20 MPa. However, these bodies lost their mechanical strength at an early stage of the immersion showing compression strength of only 7–8 MPa at 14 days of immersion. The initial strength of composite glass bodies (7–11 MPa) was lower compared with bodies containing only one kind of glass but the bodies showed no notable mechanical weakening during the test. Softening of the surface of smooth bioactive glass plates correlated well with the formation of the silica-rich layer on the plate. Interestingly, the study also showed that in porous glass structures containing both bioactive and biocompatible glass the biocompatible glass can act as a site for calcium phosphate precipitation. © 2000 Elsevier Science B.V. All rights reserved.

1. Introduction

When in contact with body fluids or tissues, bioactive glasses develop reactive layers at their surfaces resulting in chemical bond between implant and host tissue [1]. Unlike most other

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bioactive materials, the rate of chemical reactions of bioactive glasses can easily be controlled by changing the chemical composition of the glass [2]. This property of bioactive glasses makes them promising in particular for clinical applications. In fact, bioactive glasses have already been used clinically for several years, for example to replace damaged parts of facial injuries, replacement of the small bones (ossicles) in the middle ear and in surgery to fill defects in bone [3–6]. Unfortunately, the range of hot working of the bioactive glasses has been narrow. Reheating of the material has resulted in a decrease of the bioactivity [7] probably because of devitrification of the glass. This property of bioactive glasses has limited the clinical use of the material and bioactive glass implants have been applied mainly as crushed glass or non-porous rigid implants. As such, bioactive glasses represent a potential option within bioactive materials and several groups have recently focused on the development of methods for making, for instance, porous bioactive glasses [8–11]. Porosity of the bioactive glass does not only noticeably increase the total reacting surface of the glass but also allows a three-dimensional formation of the healing bone tissue.

A series of novel bioactive glasses was introduced during the mid 90s. These glasses can be used in different reheating processes without risk of decrease of the bioactivity [12]. The possibility of shaping these novel bioactive glasses has broadened the range of manufacturing methods for making, for example rigid porous bioactive glass textures with a controlled macro-porosity. As a first step toward possible clinical application of these novel bioactive glasses, the present study examines the structural strength of and the reactions occurring inside rigid porous bioactive glass

bodies immersed in a simulated body fluid (SBF) [13] as comparison with non-porous glass rods. In order to be bioactive, the silica content of the glass must be between 45–55 wt% [1,14]. In the current work, we also studied glasses containing silica >58 wt% (here defined as biocompatible glasses). The additional aim of the work was to study the chemical reactions occurring on biocompatible glass spheres in porous composite matrixes made by mixing biocompatible glass spheres with bioactive glass spheres. Finally, stiffness (Young's modulus) of implant material has a major impact on the performance of the implant-host bone interface [15]. In this study, we also measured the stiffness of the bioactive glass surface as a function of time of immersion in SBF.

2. Experimental

2.1. Glass material

Three glasses were used to make four different porous glass bodies. The composition of the studied glasses is given in Table 1. The glasses were made by mixing analytical grade Na_2CO_3 , K_2CO_3 , MgO , CaCO_3 , H_3BO_3 and $\text{CaHPO}_4 \cdot 2\text{H}_2\text{O}$ and commercial Belgian quartz sand and melting the batches in a Pt-crucible at 1360°C for 3 h. The glasses were cast, annealed, crushed and remelted to improve homogeneity. After recasting and crushing the glass particles were spherodized by feeding them into acetylene–oxygen flame [17]. In order to obtain a controlled porosity of the bodies, only a narrow fraction of microspheres (diameter 250–300 μm) was used. For sintering, 14-mm deep holes were drilled into a graphite mold using a 4-mm conical drill bit. Then, the holes were filled

Table 1
Glass compositions in weight percentage (wt%)

Glass	Na_2O	K_2O	MgO	CaO	B_2O_3	P_2O_5	SiO_2
Bioactive ^a	6	12	5	20	–	4	53
Biocompatible 1 ^b	25.5	–	–	11	1.3	2.5	59.7
Biocompatible 2	16.6	5.0	10.3	6.6	1.0	2.0	58.5

^a Glass 13–93 Brink et al. [12]

^b Glass S59, 7P2, 5 Lindfors et al. [16]

with glass microspheres of the narrow fraction and sintered to porous conical pieces (length 14 mm, top diameters of 4 and 2.8 mm, respectively) (Fig. 1). Four different types of cones were made using (A) only bioactive glass, (B) a blend of 35% spheres of the bioactive glass and 65% of the slow reacting glass 1 (biocompatible glass 1), (C) a blend of 35% of the bioactive glass and 65% of the slow reacting glass 2 (biocompatible glass 2) and (D) only the biocompatible glass 2 (Table 2). The blends and the sintering times are given in Table 2. The sintering temperature and times used for the glass spheres were chosen after testing of several combinations. Lower temperature will prolong the sintering time and increase the risk for devitrification of the glass. Higher temperatures will shorten the sintering time and, consequently, complicate the control of the appropriate extent of sintering. After sintering the test cones had the same porosity of about 30%. The final test bodies were obtained by cutting the cones from the middle to 4 mm long pieces with base and top diameters of 3.2 and 3.0 mm, respectively.

The bodies were immersed in SBF for 3, 7, 14, 21 and 42 days at $37 \pm 0.5^\circ\text{C}$ without stirring the solution. At each time interval, the concentration of Ca^{2+} and SiO_4^{4-} ions of the immersion solutions, the mechanical strength of the test bodies and the thickness of the silica-rich reaction layer formed on the bioactive glass microspheres were measured. For the analysis, four glass bodies of the same test body category were placed in a poly-

styrene container and 28 ml of SBF was added into each container giving the approximate glass surface area (SA) to SBF volume (V) ratio, $\text{SA}/V = 0.5 \text{ cm}^{-1}$.

Thus, totally five containers, one for each time interval, were needed for each category (A, B, C, and D). Then at each time interval, the four test bodies of each glass category were removed from the containers and kept in ethanol for further analysis. The quantitative analysis of Ca^{2+} and SiO_4^{4-} ions of each immersion solution was performed using a direct current plasma atomic emission spectrometer (DCPAES).¹ Wavelengths recommended by the instrument manufacturer were used for the determinations. Errors due to ionization effects caused by high-sodium concentrations in the sample solutions were eliminated by using the same sodium concentrations also in the calibration solutions (125, 200, and 250 ppm). The method was also tested by analyzing the used SBF and the agreement was good. The precision of the method, expressed as a relative S.D., is about 1–2% and the relative analytical error slightly higher. One measurement was performed on each concentration of the immersion solutions.

Thereafter, three of the four test bodies of each container were randomly selected to measure the mechanical strength of the porous bodies. For the measurements a compression-testing device² equipped with a control unit was used. The load was recorded by a load cell with a precision of $\pm 0.1 \text{ N}$. The rate of compression was 0.5 mm/min. The load was applied by a cylindrical steel plug with the load head of 6.16 mm². The maximal compression load at failure was obtained from the recorded load–deformation curves.

To measure the thickness of the silica-rich layer formed on the bioactive glass microspheres, the fourth sample of each container was studied by means of back-scattered electron imaging of scanning electron microscopy equipped with energy dispersive X-ray analysis (BEI-SEM/EDXA). For the study, the porous glass body was embedded in isobornylmethacrylate. After hardening of

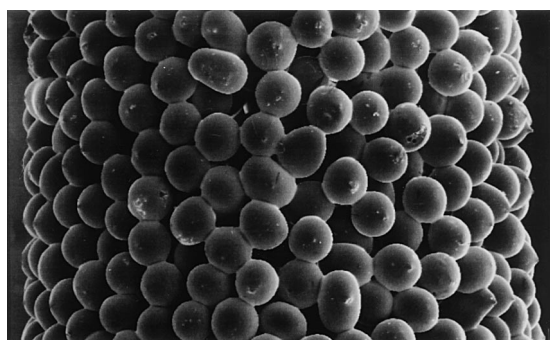


Fig. 1. SEM micrograph showing surface of a porous cone made by sintering bioactive glass microspheres of diameter 250–300 μm (original magnification $\times 36$).

¹ SpectraSpan III B, Spectrametric Inc. Andover, MA, USA.

² Alwetron TCT 20.

Table 2

Glass blends of the tested body types and their sintering times at 760°C

Body type	Bioactive (%)	Biocompatible 1 (%)	Biocompatible 2 (%)	Time (s)
A	100	–	–	360
B	35	65	–	225
C	35	–	65	240
D	–	–	100	240

the acrylate, the specimens were ground in the cross-sectional plane perpendicular to the long axes of the body until section area in the middle of each body was reached. The polished surface of each specimen was carbon coated with a sputter coater. To identify the silica-rich reaction layers formed on the glass spheres a SEM equipped with BSE detector was used. Micrographs of each specimen were recorded using a video copy processor. To compare the rate of chemical reactions occurring on glass microspheres with that on a non-porous flat surface of the same glass, the thickness of the silica-rich gel layer was measured on eight randomly selected glass spheres of the bioactive glass body at each time interval.

To study the formation rate and the stiffness of the silica-rich gel layer on a non-porous flat bioactive glass, a parallel immersion experiment was carried out. For this experiment, five bioactive non-porous glass rods with rectangular cross-section ($10 \times 14 \times 40 \text{ mm}^3$) were cast using the same glass composition as in the bioactive test bodies A. To measure the stiffness of the intact glass surface, an indentation test was performed by loading the area of interest under axial compression using a cylindrical indentator connected to a force transducer.³ The rate of indentation was $3.75 \text{ } \mu\text{m/s}$. To avoid artifacts caused by unevenness of the measured surface, the area of interest was selected from the uppermost surface of the glass rod. Due to the surface tension, the uppermost free surface of a cast glass rod is the smoothest. In order to apply an even compressive stress on the thin and flexible layer of silica-rich gel at measurements performed at each time interval of the immersion, a plane-ended indentator, 0.8 mm in diameter was used. After measuring the initial hardness of the glass

surface each was placed in a separate container, one for each time interval (3, 7, 14, 21, and 42 days). SBF was added to obtain the same SA/V ratio as used in immersion of the porous bodies. At each time interval, the stiffness of the glass surface was analyzed after gentle rinsing with ethanol and drying. Although the plane end gives artificial stress gradients around its periphery [18] the measurement results demonstrated changes in surface stiffness. Further, the thickness of silica gel on each sample was measured by means of BEI-SEM. All measurements were performed on eight randomly selected locations of each bioactive glass rod.

The group statistics of the measurement results were calculated using analysis of variance (ANOVA) with Tukey's post-hoc *t*-test. Paired comparisons were performed using paired students *t*-test. For all statistical tests, significance was defined as $P < 0.05$.

3. Results

The concentration of Ca^{2+} in the immersion solutions increased following the break down of the silica network and showed the maximum at three days (Fig. 2((a) and (b))). The solubility of Ca^{2+} and SiO_4^{4-} into SBF was highest from bodies containing only bioactive glass microspheres. The rate of leaching was lower from the composite cones and lowest from the bodies sintered from spheres made of biocompatible glass 2. After three days, the dissolution of Ca^{2+} was slowed down by the formation of silica-rich layer on the glass microspheres (Fig. 2((a) and (b))).

Porous bodies made by sintering only one type of glass microspheres showed highest values of initial compression strength (17.4–20.0 MPa) (Table 3). However, porous cones of type A containing only bioactive glass microspheres lost their mechanical strength significantly ($P < 0.05$) dur-

³ RTS-1000, Transducer Techniques, Temecula, USA.

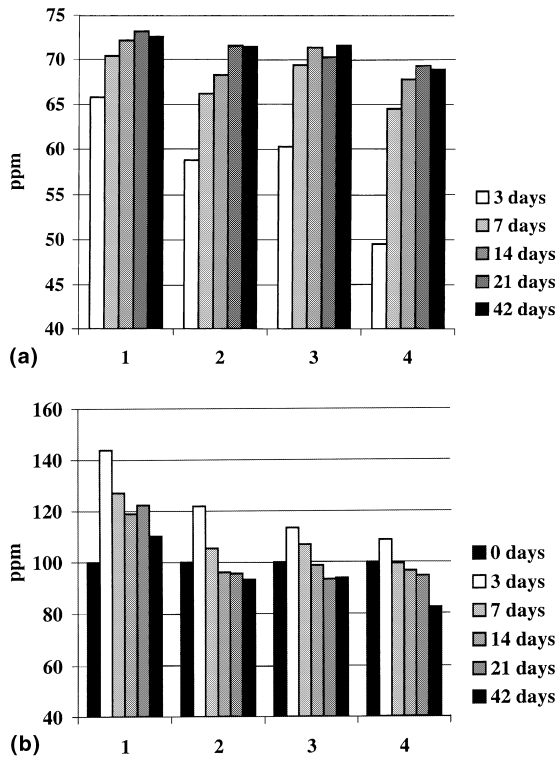


Fig. 2. Increasing concentrations of SiO_4^{4-} (a) and decreasing concentrations of Ca^{2+} (b) in the immersion solutions as a function of immersion time (3–42 days). Note that the initial concentration of Ca^{2+} in SBF is close to 100 ppm. The bars represent the SiO_4^{4-} or Ca^{2+} concentrations of the immersion solutions in one container containing four glass samples. A = bodies made of only bioactive glass spheres, B = bodies made of a blend containing 35% of the bioactive glass and 65% of the biocompatible glass 1 spheres, C = bodies made of a blend containing 35% of the bioactive glass and 65% of the biocompatible glass 2 spheres, D = bodies made of only the biocompatible glass 2 spheres.

ing the first 14 days of immersion. Thereafter, simultaneously with the calcium phosphate precipitation, there was a trend for increased compression strength of the bodies. Composite bodies containing both bioactive and biocompatible glass spheres (bodies B–C) showed only minor changes in compression strength during immersion in SBF. The strength of sintered cones of type D containing only spheres made of the biocompatible glass 2 was decreased significantly within one week during the immersion.

BEI-SEM/EDXA analysis showed that there was a significant difference ($P < 0.001$) in the average thickness of the silica-rich gel layer formed on bioactive microspheres compared with that formed on the flat surface of bioactive glass rods (Fig. 3). The silica-rich gel layer formation

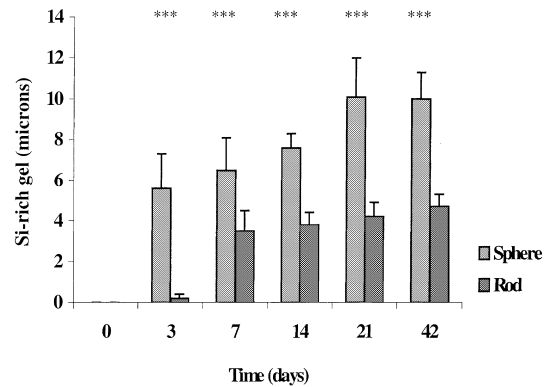


Fig. 3. Thickness of the silica-rich layer on bioactive microspheres and on glass rod as a function of immersion time. Statistical significance of the differences between the layer thickness indicated by *** ($P < 0.001$).

Table 3

Compression strengths of the tested body types as a function of immersion time in SBF^a

Body type	Intact (MPa)	3 + 7 days (MPa)	14 days (MPa)	21 + 42 days (MPa)
A	20.0 ± 0.7	13.5 ± 7.8 ^b	6.7 ± 1.3	11.0 ± 4.6
B	10.9 ± 4.1	9.1 ± 3.1	9.0 ± 1.8	8.1 ± 3.5
C	7.4 ± 4.6	11.7 ± 5.2	10.4 ± 1.8	6.5 ± 3.2
D	17.4 ± 5.6 ^b	10.4 ± 3.5 ^b	7.8 ± 0.6	7.4 ± 2.4 ^c

^a The values represent the means ± S.D. ($n = 3-6$). Statistical significance of the time-related changes indicated by superscripts (analysis of variance with Tukey's post-hoc test).

^b $P < 0.05$.

^c $P < 0.01$ compared with intact cones.

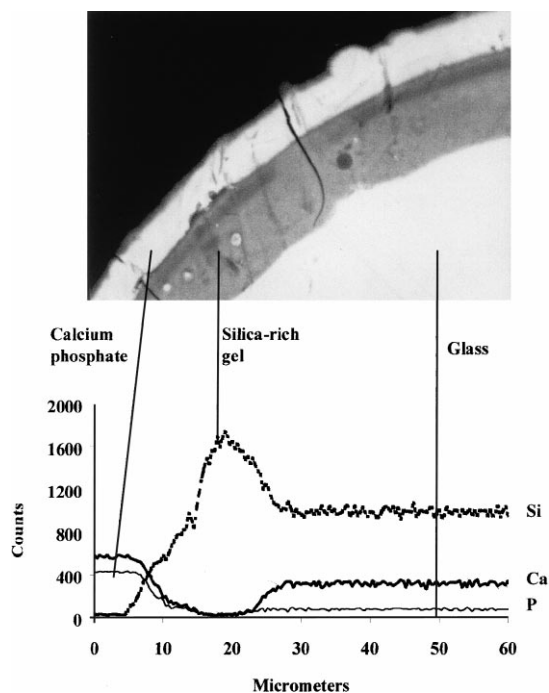


Fig. 4. EDXA line-profiles of Si, Ca, and P scanned over a cross-section of a bioactive glass sphere at 21 days of immersion in SBF. The line-profiles demonstrate bulk glass, silica-rich layer, and calcium phosphate layer (original magnification $\times 1700$).

occurred on bioactive microspheres already at three days, while formation of a comparable gel layer on a non-porous flat surface of the same glass was not obtained until at seven days. In both instances, the silica-rich gel layer was finally covered by calcium phosphate layer (Fig. 4).

The hardness of the flat bioactive glass surface decreased during the first 21 days of immersion (Fig. 5). Thereafter, there was a trend for an increased hardness of the surface during the formation of calcium phosphate layer.

4. Discussion

In order to be bioactive, the silica content of the glass must be between 45–55 wt% [1,14]. The lower the silica content the easier it is dissolved in the surrounding body fluid. In the current study, the rate of dissolution of Ca^{2+} and SiO_4^{4-} was highest

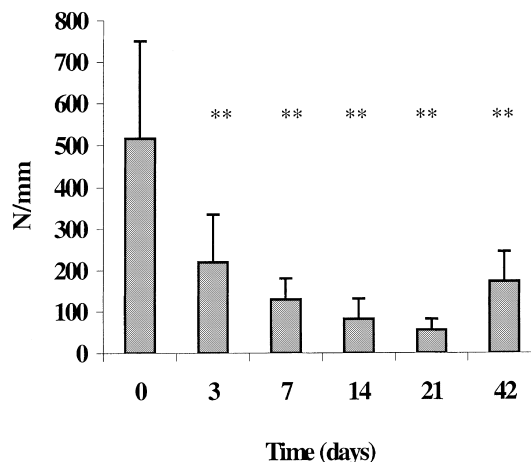


Fig. 5. Surface hardness of bioactive glass rod as a function of immersion time. The bars represent the hardness values of the specimens ($n = 5-8$) as function of immersion time. The values (N/mm) represent mean $\pm$ S.D. Statistical significance of the differences between the layer thickness indicated by ** ($P < 0.01$).

from the bodies containing only bioactive glass microspheres. The SiO_2 content of the bioactive glass was 53 wt%. Composite bodies containing both biocompatible and bioactive glass spheres showed lower rate of dissolution. The rate was lowest for the biocompatible glass 2 containing 58.5 wt% SiO_2 . Principally, in spite of repeated melting of the glass during the spheroidization and sintering procedures, the dissolution of ions from the porous bioactive glass bodies seemed to be similar that is reported in earlier works by several investigators [18–24]. The novel bioactive glass used in the study had maintained its high level of bioactivity in spite of several reheating procedures. This can be explained by unchanged amorphous structure of the glass after working in heat (Fig. 6), while crystallization of a conventional bioactive glass [5] during reheating is clearly shown in Fig. 7. There was a significant difference in the rate of formation of the silica-rich layer on bioactive glass microspheres inside the porous bodies compared with that on the surface of the bioactive glass rod. This can probably be explained by the high surface energy of the necks combining sintered microspheres in the sintered body. As a result, a rapid dissolve of ions from the porous glass body occurs

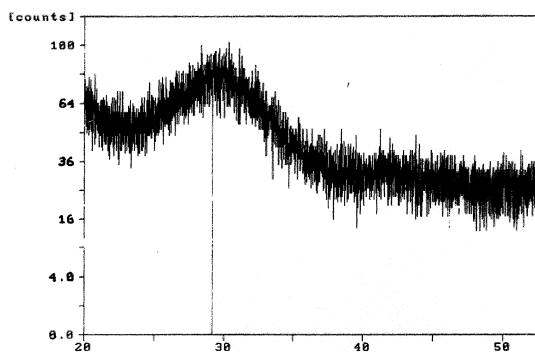


Fig. 6. X-ray diffraction patterns of pulverized porous cone made by sintering microspheres of a novel bioactive glass (glass 13–93 of Brink et al. [12]). The microspheres were manufactured using flame-spraying methods described by Pitkänen et al. [17]. Sintering of the spheres was performed in a graphite mold at 760°C for 360 s. Smooth diffraction pattern of the glass powder indicates no phase transformation and totally glassy phase [8].

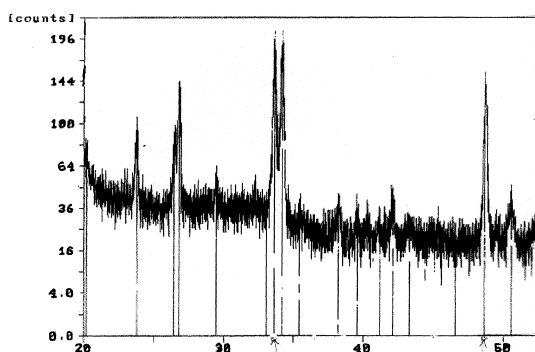


Fig. 7. X-ray diffraction patterns of pulverized porous cone made by sintering microspheres of a conventional bioactive glass [5]. The microspheres were manufactured using flame-spraying methods described by Pitkänen et al. [17]. Sintering of the spheres was performed in a graphite mold at 760°C for 360 s. High peaks of the diffraction pattern indicate devitrification and a phase transformation of the glass [8].

and the glass surface is fast depleted leaving a silica-rich gel (Fig. 8). Decreased mechanical strength of the bioactive glass bodies during the immersion speaks also for this interpretation: necks combining the microspheres dissolve and the structure is weakened (Fig. 9). In composite bodies, 2/3 of the microspheres consisted of a slowly dissolving biocompatible glass. Consequently, the formation of a silica-rich gel layer is slow on necks and spheres of 2/3 of the sintered structure. The

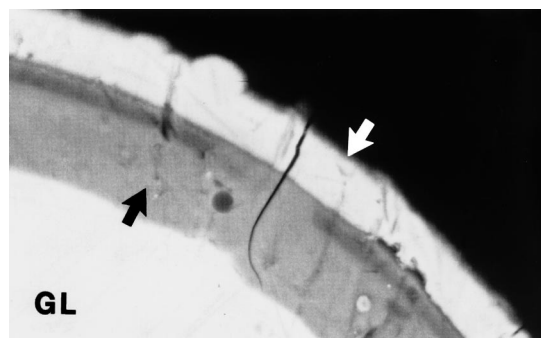


Fig. 8. BEI-SEM micrograph showing reaction layers formed on a bioactive glass microsphere at 21 days. According to EDXA, the gray layer on the intact glass (GL) is silica-rich (black arrow) and the uppermost white layer (white arrow) on the silica-rich layer is calcium phosphate (original magnification $\times 1700$).

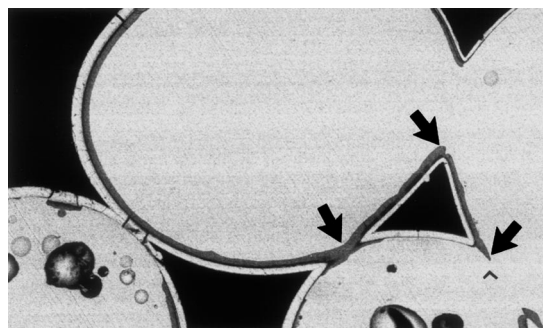


Fig. 9. BEI-SEM micrograph showing dissolving necks connecting bioactive microspheres at 14 days of immersion. Dissolution of the glass and formation of the silica-rich gel (black arrows) results in weakening of the sintered glass structure (original magnification $\times 300$).

lower initial mechanical strength of the composite bodies can be explained by different coefficients of thermal expansion of the glasses resulting in micro cracks in the sintered glass body. In spite of the lowest rate of dissolution of ions from the bodies sintered only of the biocompatible glass 2 spheres, their mechanical strength was weakened fast. These two results are contradictory.

The role of biocompatible glass spheres mixed with highly bioactive spheres in a porous glass body can be, however, of interest. The following experiment was monitored to study this. One

bioactive particle was placed in the middle of slow reacting biocompatible glass spheres and the lot was sintered to a porous body (Fig. 10). In general, the biocompatible glass spheres were covered only by a thin layer of silica gel, while calcium phosphate precipitated on the bioactive particle. However, the biocompatible particles sharing the porosity with the bioactive one were also covered by a calcium phosphate precipitate. The concentration of Ca^{2+} and HPO_4^{2-} in SBF is 2.5 mM and 1.0 mM, respectively [13]. Consequently, $[\text{Ca}^{2+}][\text{HPO}_4^{2-}] = 2.5 \times 10^{-6}$, i.e. very close to the solubility product of calcium monohydrate phosphate ($L_{\text{CaHPO}_4} = 3.4 \times 10^{-6}$ at 38°C) [25]. Thus, the concentration of calcium phosphate in SBF is very close to saturation. Migration of Ca^{2+} and PO_4^{3-} ions from the bioactive glass particle through the silica gel layer into the SBF further increases the concentrations. In the vicinity of the bioactive particle, the solubility product of calcium phosphate is easily exceeded resulting in calcium phosphate precipitation not only on the bioactive glass but also on the neighboring biocompatible glasses. The gel probably acts as nucleation site.

Bioactive glass particles of a narrow fraction has been studied extensively as fillers of bone defects [26–31]. The filled defects have shown a fairly

rapid growth of new bone. This study showed that by sintering bioactive glass microspheres, it is possible to obtain rigid porous textures in which the whole bioactive glass surface of the three-dimensional porous structure reacts fast in SBF. According to Hench [32], the faster the surface of a bioactive glass implant reacts, the faster is bonding of the glass implant to host bone. Porous bioactive glass bodies allow the reactions to occur three-dimensionally resulting in numerous centers of bioactivity [33], suggesting that also the rate of new bone formation *in vivo* is high.

Further, the major problem in orthopedics is loosening of hip prostheses. One of the main reasons for this is the postoperative micro-motion of the stem [15]. Bioactive glasses are fixed to the host bone by a chemical bond to the calcium phosphate layer formed on the bioactive glass. However, between the phosphate layer and the glass there is a silica-rich gel layer. The results of our study showed that this silica gel layer is flexible suggesting that the gel layer could act as a buffer against the micro-motion. Finally, using of porous bioactive glass textures as an implant coating may provide means to guide and promote the ingrowth of new bone into selected anchoring areas of cementless implants.

5. Conclusions

Spherodization and sintering of the bioactive glass did not affect the bioactivity as measured by formation of the silica-rich reaction layer on the glass during an *in vitro* immersion. In porous bioactive glass bodies the rate of formation of silica-rich layer on the bioactive glass spheres was double compared with the corresponding formation on a flat surface of the same glass. The net bioactivity of porous glass bodies can be controlled by mixing bioactive and biocompatible microspheres and sintering the mix.

Acknowledgements

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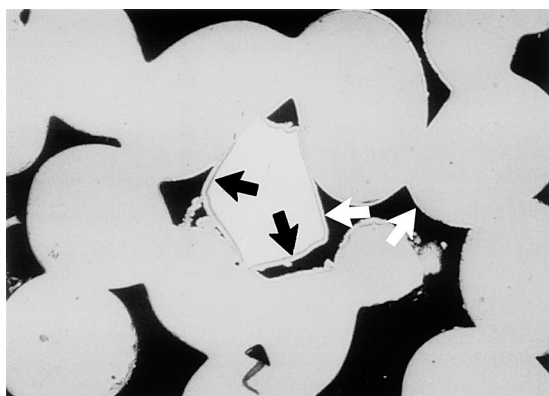


Fig. 10. BEI-SEM micrograph showing bioactive glass particle (white) in the middle of a porous body of sintered biocompatible microspheres (gray). According to EDXA analysis, the dark layer (black arrows) on the bioactive particle demonstrates silica-rich layer, and the white layer (white arrows) on the silica-rich layer, as well as on neighboring compatible glasses, is calcium phosphate (original magnification $\times 100$).

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References

- [1] L.L. Hench, Ö. Andersson, *An Introduction to Bioceramics*, World Scientific, Singapore, 1, 1993 p. 45.
- [2] K.H. Karlsson, M. Rönnlöf, *Glastech. Ber., Glass Sci. Technol.* 71 (1998) 141.
- [3] J. Wilson, A. Yli-Urpo, R-P. Happonen, *An Introduction to Bioceramics*, World Scientific, Singapore, 1, 1993 p. 65.
- [4] K. Aitasalo, J. Suonpää, M. Peltola, A. Yli-Urpo, *Bioceramics*, Pergamon, Oxford, 10, 1997 p. 429.
- [5] J.T. Heikkilä, K.T. Mattila, Ö.H. Andersson, J. Knuuti, A. Yli-Urpo, A.J. Aho, *Bioceramics*, Pergamon, Oxford, 8, 1995 p. 35.
- [6] J.T. Heikkilä, H.J. Aho, A. Yli-Urpo, R-P. Happonen, A.J. Aho, *Acta Orthop. Scand.* 66 (1995) 463.
- [7] O. Peitl Filho, G.P. LaTorre, L.L. Hench, *J. Biomed. Mater. Res.* 30 (1996) 509.
- [8] F-H. Lin, C-C Lin, H-C Liu, Y-Y Huang, C-Y Wang, C-M Lu, *Biomaterials* 15 (1994) 1087.
- [9] A. El-Ghannam, P. Ducheyne, I.M. Shapiro, *Biomaterials* 18 (1997) 295.
- [10] A. El-Ghannam, P. Ducheyne, I.M. Shapiro, *J. Biomed. Mater. Res.* 36 (1997) 167.
- [11] H.O. Ylänen, T. Helminen, A. Helminen, J. Rantakokko, K.H. Karlsson, H.T. Aro, *Ann. Chi. Gynaecol.* 88 (1999) 237.
- [12] M. Brink, V. Pitkänen, J. Tikkanen, M. Paajanen, G. Graeffe, *Bioceramics*, Pergamon, Oxford, 9, 1996 p. 127.
- [13] T. Kokubo, H. Kushitani, S. Sakka, T. Kitsugi, T. Yamamuro, *J. Biomed. Mater. Res.* (1990) 721.
- [14] L.L. Hench, *J. Am. Ceram. Soc.* 81 (1998) 1705.
- [15] R. Huiskes, *Acta Orthop. Scand.* 64 (1993) 699.
- [16] N. Lindfors, P. Slätis, K.H. Karlsson, Ö.H. Andersson, *Bioceramics*, Pergamon, Oxford, 8, 1995 p. 107.
- [17] V. Pitkänen, M. Tikkanen, G. Graeffe, M. Brink, K.H. Karlsson, *J. Aerosol Sci.* 26 (1995) 831.
- [18] E.M.H. Obeid, M.A. Adams, J.H. Newman, *J. Bone J. Surg [Br]* 76B (1994) 315.
- [19] M. Ogino, F. Ohuchi, L.L. Hench, *J. Biomed. Mater. Res.* 14 (1980) 55.
- [20] M.R. Filgueiras, G. LaTorre, L.L. Hench, *J. Biomed. Mater. Res.* 27 (1993) 445.
- [21] I.M. Kangasniemi, E. Vedel, J. de Blick-Hogerworst, A.U. Yli-Urpo, K. de Groot, *J. Biomed. Mater. Res.* 27 (1993) 1225.
- [22] Ö.H. Andersson, J. Rosenqvist, K.H. Karlsson, *J. Biomed. Mater. Res.* 27 (1993) 941.
- [23] J.D. Marshall, J.Y. Josefowicz, P. Ducheyne, *Bioceramics*, Butterworth, Oxford, 6, 1995 p. 389.
- [24] S. Radin, P. Ducheyne, S. Falaize, A. Hammond, *Bioceramics*, Pergamon, Oxford, 10, 1997 p. 45.
- [25] A. Seidell, *Solubilities of Inorganic and Metal Organic Compounds*, Van Nostrand, New York, 1940 p. 320.
- [26] E.J. Schepers, P. Ducheyne, L. Barbier, S. Schepers, *Implant Dent.* 2 (1993) 151.
- [27] A.J. Garcia, P. Ducheyne, *J. Biomed. Mater. Res.* 28 (1994) 947.
- [28] D.P. Ducheyne, A. El-Ghannam, I. Shapiro, *J. Cell. Biochem.* 56 (1994) 162.
- [29] L. Wheeler, K.E. Stokes, R.G. Hoellrich, D.L. Chamberland, S.W. McLoughlin, *J. Biomed. Mater. Res.* 41 (1998) 527.
- [30] E. Schepers, L. Barbier, P. Ducheyne, *Int. J. Oral Maxillofac. Implants* 13 (1998) 655.
- [31] H. Oonishi, L.L. Hench, J. Wilson, F. Sugihara, E. Tsuji, S. Kushitani, H. Iwaki, *J. Biomed. Mater. Res.* 44 (1999) 31.
- [32] L.L. Hench, J.K. West, *Life Chem. Rep.* 13 (1996) 187.
- [33] E.J. Schepers, P. Ducheyne, *J. Oral Rehabil.* 24 (1997) 171.